

9(a)	constant voltage	M1
	that produces/dissipates same power as (the mean power of) the alternating voltage	A1
9(b)(i)	(maximum) rate of cutting of (magnetic) flux doubles	B1
	(peak and hence) r.m.s. induced e.m.f. doubles	B1
9(b)(ii)	sketch: (sinusoidal) wave of period 10 ms	B1
	peak E shown as $\pm 34\text{ V}$	B2
	<i>(1 mark out of 2 awarded if peak E shown as $\pm 17\text{ V}$ or $\pm 24\text{ V}$)</i>	
9(c)	current in the coil results in forces that oppose its rotation or current in the resistor dissipates the energy of rotation	B1
	coil stops rotating	B1